Chemistry Detailed Notes

WBJEE • JEE Main • NEET common Chemistry preparation

1. Some Basic Concepts of Chemistry

• Chemistry studies composition, structure, properties and transformations of matter.

• Mole concept: one mole contains Avogadro's number of particles. Use n = given mass / molar mass.

• Molar mass is the mass of one mole of a substance, numerically equal to molecular/formula mass in g mol^{-1}.

• Empirical formula gives the simplest whole-number ratio of atoms; molecular formula gives the actual ratio.

2. Atomic Structure

• Atoms contain protons, neutrons and electrons. Atomic number Z equals the number of protons.

• Mass number A = protons + neutrons. Isotopes have the same Z but different neutron numbers.

• Electron configuration follows Aufbau filling, Pauli exclusion and Hund's rule.

• Important ideas: shells, subshells, orbitals, quantum numbers and electronic configuration.

3. Periodic Classification

• Periodic trends include atomic radius, ionic radius, ionization enthalpy, electron gain tendency and electronegativity.

• Across a period effective nuclear charge generally increases; down a group the number of shells increases.

• Metallic character generally increases down a group and decreases across a period.

4. Chemical Bonding

• Ionic bonding involves electrostatic attraction between oppositely charged ions; covalent bonding involves sharing of electrons.

• VSEPR theory predicts molecular shape from repulsion among electron pairs.

• Hybridization commonly discussed at this level includes sp, sp^2 and sp^3.

• Sigma bonds result from head-on overlap; pi bonds result from sidewise overlap of suitable orbitals.

• Polarity depends on electronegativity difference and molecular geometry.

5. States of Matter & Gases

• Ideal gas equation: $PV = nRT$. Always use absolute temperature in gas-law calculations.

• Boyle's law relates pressure and volume at constant temperature; Charles' law relates volume and temperature at constant pressure.

• Real gases deviate from ideal behavior especially at high pressure and low temperature.

6. Thermodynamics

• System, surroundings, state functions, heat and work are basic thermodynamic ideas.

• Enthalpy change represents heat change at constant pressure for common reaction situations.

• Exothermic processes release heat; endothermic processes absorb heat.

• Hess's law allows reaction enthalpy to be obtained by combining known thermochemical equations.

7. Equilibrium

• Chemical equilibrium is dynamic: forward and reverse processes continue while macroscopic composition remains constant.

• Le Chatelier's principle predicts the direction of shift after a change in concentration, pressure or temperature.

• Equilibrium constants describe the relative amounts of products and reactants at equilibrium.

8. Solutions

• Molarity M = moles of solute / volume of solution in litres.

• Dilution keeps the amount of solute constant, giving $M_1V_1 = M_2V_2$.

• Concentration units and colligative-property concepts are important for numerical practice.

9. Redox Reactions & Electrochemistry

• Oxidation is loss of electrons/increase in oxidation state; reduction is gain of electrons/decrease in oxidation state.

• Oxidation number rules help identify electron-transfer changes.

• Electrochemical cells convert chemical and electrical energy; anode is the site of oxidation and cathode is the site of reduction.

10. Organic Chemistry Foundations

• Understand functional groups, nomenclature, isomerism, electronic effects and common reaction types.

• Inductive effect, resonance and hyperconjugation help explain stability and reactivity.

• Acid-base behavior and reaction intermediates should be learned together with representative examples.

11. Hydrocarbons

• Alkanes mainly undergo substitution; alkenes and alkynes commonly undergo addition reactions.

• Markovnikov-type orientation and basic mechanisms are useful for solving reaction-based questions.

• Aromaticity and electrophilic substitution are key ideas for benzene chemistry.

12. Practical Revision Checklist

- Revise definitions first, then reactions and trends, then numerical formulas.
- Make a one-page list of exceptions and frequently confused concepts.
- After every chapter, solve mixed questions without looking at notes.